

Xia Su

AI that bridges people and spaces.

xiasu@cs.washington.edu | xiasu.github.io | [Google Scholar](#)

I study how AI can help people understand, navigate, and shape the spaces around them. My work connects architecture, human-computer interaction, accessibility, and computer vision. At Google, I explore how vision algorithms and AI agents can work across devices and the cloud to understand environments, human behavior, and the relationships between them.

EXPERIENCE

Google

Aug 2026 - Present

Software Engineer | Computer vision and spatial AI

- Research collaborative on-device and cloud deployment of computer vision algorithms and AI agents under resource constraints.
- Explore multi-level understanding of spaces and human behavior, connecting environmental context with how people interact with their surroundings.
- Investigate context-sensitive safety-risk alerts that respond to user needs and available computational resources.

University of Washington

Sep 2021 - Aug 2026

Graduate Research Assistant / Teaching Assistant

Apple AIML

Apr - Sep 2025

Machine Learning Intern

Adobe Research

Jun - Sep 2024; Jun - Sep 2023

Research Scientist Intern

Microsoft Research Asia

Jun 2020 - Mar 2021

Research Intern

EDUCATION

University of Washington

2021 - 2026

Doctoral research, Computer Science & Engineering; dissertation defended Aug 2026.

M.S., Computer Science & Engineering, 2024. Advisor: Jon E. Froehlich.

Tsinghua University

2014 - 2021

Bachelor of Architecture, 2014 - 2018; master's degree, 2018 - 2021.

Research in architecture, computational design, and indoor scene generation.

TECHNICAL EXPERTISE

Vision & spatial computing. Dense prediction and segmentation; visual SLAM; structure from motion; 3D reconstruction and scene synthesis; mobile AR; CAD and fabrication.

AI & agent systems. PyTorch; transformers and ViTs; SFT, PEFT, and GRPO; RAG; agent workflows; LangGraph; Model Context Protocol; custom objectives and evaluation.

Systems & interaction. Full-stack web development; cross-platform mobile/AR applications; robotics and embedded integration; real-time graphics; UI/UX design.

Research & evaluation. Benchmark and dataset design; data curation; quantitative and qualitative analysis; human-centered AI; user studies and statistical analysis.

SELECTED FULL PAPERS

CapNav: Benchmarking Vision Language Models on Capability-conditioned Indoor Navigation

Xia Su, Ruiqi Chen, Benlin Liu, Jingwei Ma, Zonglin Di, Ranjay Krishna, Jon E. Froehlich

[CVPR 2026](#)

DepthScape: Authoring 2.5D Designs via Depth Estimation, Semantic Understanding, and Geometry Extraction

Xia Su, Cuong Nguyen, Matheus A. Gadelha, Jon E. Froehlich

[DIS 2026](#)

FlyMeThrough: Human-AI Collaborative 3D Indoor Mapping with Commodity Drones

Xia Su, Ruiqi Chen, Jingwei Ma, Chu Li, Jon E. Froehlich

[UIST 2025](#)

AccessibilityScout: Personalized Accessibility Scans of Built Environments

William Huang, Xia Su, Jon E. Froehlich, Yang Zhang

[UIST 2025](#)

SonifyAR: Context-aware Sound Generation in Augmented Reality

Xia Su, Jon E. Froehlich, Eunye Koh, Chang Xiao

[UIST 2024](#)

RASSAR: Room Accessibility and Safety Scanning in Augmented Reality

Xia Su, Han Zhang, Kaiming Cheng, Jaewook Lee, Qiaochu Liu, Wyatt Olson, Jon E. Froehlich

[CHI 2024](#)

Kinergy: Creating 3D Printable Motion using Embedded Kinetic Energy

Liang He, Xia Su, Huaishu Peng, Jeffrey I. Lipton, Jon E. Froehlich

[UIST 2022](#)

Interior Layout Generation Based on Scene Graph and Graph Generation Model

Xia Su, Chenglin Wu, Wen Gao, Weixin Huang

[Design Computing and Cognition '20](#), pp. 267-282

SELECTED POSTERS & DEMOS

Authoring 2.5D Designs with Depth Estimation

Xia Su, Cuong Nguyen, Matheus A. Gadelha, Yu Shen, Stefano Petrangeli, Jon E. Froehlich

[CHI 2025 Late-Breaking Work](#)

A Demo of DIAM: Drone-based Indoor Accessibility Mapping

Xia Su, Ruiqi Chen, Weiye Zhang, Jingwei Ma, Jon E. Froehlich

[UIST 2024 Demo](#)

RAIS: Towards A Robotic Mapping and Assessment Tool for Indoor Accessibility Using Commodity Hardware

Xia Su, Daniel Campos Zamora, Jon E. Froehlich

[ASSETS 2024 Poster](#)

PATENTS & INVENTIONS

Context-aware Audio Data Acquisition

Chang Xiao, **Xia Su**, Eunye Koh. Adobe Inc.

Published patent application US20260023522A1, Jan 22, 2026.

FlyMeThrough: Human-AI Collaborative 3D Indoor Mapping with Commodity Drones

Xia Su, Ruiqi Chen, Jon E. Froehlich. UW CoMotion.

Authoring 2.5D Designs with Depth Estimation

Xia Su, Cuong Nguyen, Matheus A. Gadelha, Yu Shen, Stefano Petrangeli.

Adobe Research.

ACADEMIC SERVICE

Reviewer: CHI 2024, CHI 2024 Late-Breaking Work, SUI 2024, UIST 2024, CHI 2025, CHI 2026.

Reviewer: IEEE International Conference on Robotics and Automation (ICRA), 2025.

PRESS & INVITED TALKS

Using augmented reality to improve accessibility

Oregon Public Broadcasting | Nov 17, 2023

RASSAR: Room Accessibility and Safety Scanning in Augmented Reality

UW CSE Colloquium | Nov 2, 2023

AR for Accessibility and Creativity

Brown HCI Lab | Oct 26, 2023

Q&A: Researchers aim to improve accessibility with augmented reality

UW News | Oct 17, 2023

Evaluating Real-world Accessibility

Igarashi Lab, University of Tokyo | May 14, 2023

TEACHING

Teaching assistant, UW CSE 442: Data Visualization.

Teaching assistant, UW MHCID: Prototyping Studio.

Teaching assistant, UW CSE 412: Introduction to Data Visualization for non-CSE majors.

Teaching assistant, UW CSE 160: Data Programming.

Guest lecturer, UW CSEP 590B: The Future of Access Technology, Winter 2023.

Guest lecturer, UW CSE 493E: The Future of Access Technology, Fall 2023 and Fall 2024.

MENTORING

Ruiqi (Richard) Chen, master's student, University of Washington; began May 2024.

Weiye Zhang, undergraduate, University of Washington; began Mar 2024.

Qiaochu (Steve) Liu, master's student, Tsinghua University; Dec 2022 - May 2023.

Jackson Ma, undergraduate, University of Washington; Oct 2022 - Feb 2023.